

Conscious tactile perception entails distinct neural dynamics within somatosensory areas

Highlights

- Somatosensory and report-related dynamics are spatiotemporally segregated
- Early responses from SI scale with both physical and perceived stimulus intensity
- Sustained responses from parietal operculum reflect stimulus awareness
- Conscious tactile processing embraces multiple dynamics within somatosensory areas

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In brief

Albertini et al. rely on human intracranial recordings to uncover the neural basis of tactile conscious perception. They find sustained, threshold-sensitive activity in the parietal operculum as the neural correlate of tactile awareness, independent of report, and challenge the involvement of the prefrontal cortex in tactile conscious perception.



Article

Conscious tactile perception entails distinct neural dynamics within somatosensory areas

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SUMMARY

Distilling the neural correlates of consciousness (NCCs) in humans is challenging due to limitations in the spatiotemporal resolution of recording techniques and confounds related to pre- and post-perceptual processes. In this study, we leveraged the detailed insights provided by human intracortical recordings to elucidate how somatosensory responses to simple tactile stimuli vary across different stimulus intensities and reporting conditions. Among the various spatiotemporal components of somatosensory processing, we observed tonic responses in posterior perisylvian regions that exhibited all the key characteristics of somatosensory NCCs. These responses remained invariant regardless of reporting, displayed an all-or-nothing pattern at the verge of the sensory threshold, and showed the most pronounced divergence between perceived and non-perceived stimuli. Overall, our findings indicate that conscious perception of simple tactile stimuli depends on higher-order somatosensory regions and that sustained neural dynamics in these areas may serve as an organizational principle of somatosensory awareness.

INTRODUCTION

A simple tactile stimulus triggers a cascade of neural dynamics within the human somatosensory system,^{1–4} whose richness has been recently revealed via intracerebral recordings.⁵ Early phasic activities in response to hand stimulation arise at approximately 20-ms post stimulus in primary somatosensory (SI), primary motor, and parietal cortices; prolonged responses emerge at approximately 30 ms in premotor areas, whereas late tonic responses, peaking at approximately 100 ms, characterize perisylvian regions. These distinct patterns were ascribed to different functions. Phasic responses in SI have been associated with basic stimulus processing,⁶ paralleling the scaling with stimulus intensity observed in its responses^{7,8} and early magneto- and electroencephalography (M/EEG) components.^{9–11} Prolonged responses have been associated with the integration of sensorimotor information from different body parts,¹² a property known to characterize premotor neurons.¹³ Finally, based on the notion that tactile awareness deficits derive from lesions in posterior perisylvian regions,^{14,15} tonic responses have been proposed to index when tactile stimuli are consciously perceived,¹² a

hypothesis further supported by the features of parietal-opercular blood-oxygen-level-dependent (BOLD) activations in tactile paradigms.^{8,16} In parallel, there exists a temporal convergence between tonic responses and the tactile awareness biomarkers derived from M/EEG studies, wherein components around 140 ms after the stimulation exhibit the largest separation between perceived and unperceived stimuli.^{9–11,17–20}

To deepen the relationship between perisylvian tonic activity and tactile awareness, we recently compared the features of tonic responses with the lesional mapping of patients exhibiting deficits in tactile perception. Our findings revealed an overlap in the parietal operculum and demonstrated that enhanced tonic responses are associated with improved detection of visuo-tactile stimuli,²¹ corroborating the hypothesis that tonic responses in posterior perisylvian regions fulfill the criteria for a neural correlate of conscious somatosensory perception.¹² Initially defined as the “minimal set of neural events jointly sufficient for a specific conscious experience,²²” a content-specific neural correlate of consciousness (NCC) must (1) occur in response to feature-specific stimuli, (2) dynamically change between perceived and unperceived contents, and (3) be



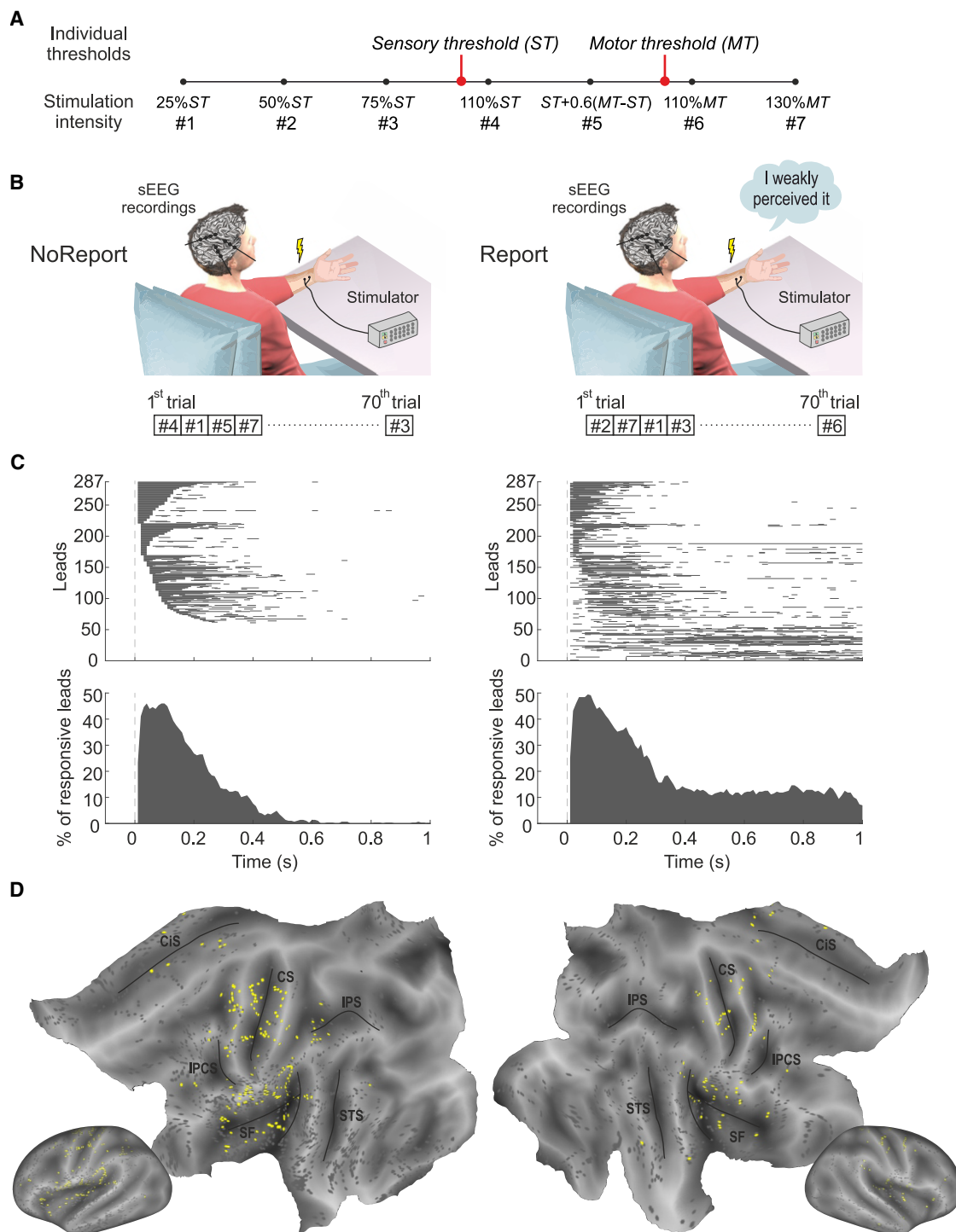


Figure 1. Intracortical responsiveness to a no-report/report tactile stimulation paradigm

(A) sEEG-recorded patients underwent stimulation of the median nerve contralateral to the recording implant at seven different intensities, derived from individual sensory and MTs, in two different conditions/blocks (see STAR Methods; Tables S1 and S2).

(B) In the first block (no-report), patients passively received the stimulations, whereas in the second block (report), after each stimulation, the patients were asked to verbally report whether they perceived it and, if so, whether the sensation was *weak* or *strong*.

(C) Distribution of significant bins for each responsive lead in NC (left) and RC (right), pooled across intensities. Leads are sorted according to their first significant bin in NC.

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insensitive to task manipulations intended at modulating the pre-requisites or consequences of consciousness.^{23–25} To date, tonic responses have only been proven to meet the first NCC requirement, with promising yet inconclusive hints about their involvement in determining conscious somatosensory perception. To fill this gap, experimental and methodological challenges must be carefully considered.

Identifying a stimulus as perceived typically implies that participants report their subjective experience. Although seemingly trivial, this process has been consistently shown to confound the NCCs, with neural activities related to expectation, memory, attention, and task relevance,^{24,26} along with motor preparation and execution.^{27,28} Distilling the NCCs from pre- and post-perceptual factors has significantly benefited from the introduction of no-report paradigms,²⁸ wherein participants are not required to provide explicit judgments regarding their perception. For example, the adoption of no-report paradigms contributed to de-emphasizing the importance of the late, fronto-central electroencephalogram (EEG) component p300^{29–33} (see also Förster et al.³⁴), long considered a signature of stimulus-conscious perception.^{35–37} No-report studies have also rekindled interest in processes occurring in sensory areas,^{8,38–41} independent of attention,⁴² expectation,⁴³ working memory,⁴⁴ and consciousness level.⁴⁵ Yet, the debate on the localization of the neural correlates of perceptual awareness^{46–48} is still ongoing, and prefrontal activations have been reported both in humans^{49–51} and animal models^{52–54} even in no-report conditions.

Besides reports and task relevance, another factor that hindered the precise delineation of NCCs is limitations inherent to non-invasive recording techniques, such as the lack of spatial resolution typical of M/EEG or the reduced capacity of neuroimaging to resolve neural events occurring within a few hundred milliseconds. In this respect, stereoelectroencephalography (sEEG) offers a fine-grained picture of neural activity with millisecond resolution and potential access to any area in the dorsal, ventral, and mesial aspects of the cortex.⁵⁵ Given these premises, sEEG opens a unique window to explore NCCs in humans, as witnessed by recent studies searching for visual^{50,52,56–59} and auditory⁶⁰ NCCs.

Here, we devised a tactile stimulation paradigm to isolate neural events fulfilling the three criteria for somatosensory NCC in patients undergoing sEEG recordings. We presented patients with a simple tactile stimulus whose cortical responsiveness is well known, modulated the intensity below and above the sensory threshold (ST), and repeated the test in both no-report condition (NC) and report condition (RC). In addition to documenting how the report impacts the somatosensory processing, we distilled the report-invariant neural activities and revealed which spatiotemporal dynamics correlate with conscious somatosensory perception.

RESULTS

Task and responsiveness

A somatosensory stimulation paradigm was administered to 30 patients undergoing sEEG long-term monitoring, whose

implanted electrodes (2,550 left, 1,254 right; see Table S1) sampled most of the cortical mantle (Figure S1A). Patients received median nerve stimulation (MNS) contralateral to the implant at seven different intensities (10 trials each, random order): 25%, 50%, 75%, and 110% of the ST; ST + 0.6 (MT – ST), where MT is the motor threshold; 110% and 130% of the MT (Figure 1A). The paradigm consisted of two conditions (Figure 1B). In the first block (NC), patients were passively stimulated, whereas in the second block after each trial the patients were asked to verbally report whether they perceived the stimulation and, if so, whether the sensation was *weak* or *strong* (RC).

The signal of each lead underwent frequency decomposition to extract its gamma-band power (GBP, 50–150 Hz, see STAR Methods), deemed to represent a proxy of the spiking activity of local neuronal populations.^{61–64} All leads significantly activated by two or more of the five highest intensities across the two conditions were classified as responsive (287/3,355 = 8.6%). An examination of the time-wise distribution of significant time bins for all responsive leads (Figure 1C) suggests that the early activation component is largely common to both conditions, whereas a distinct late component emerges specifically in the RC. Anatomically, responsive leads were distributed in a sizable portion of the cortex (yellow dots in Figure 1D), the majority being located in SI, which exhibited the peak of responsiveness (Figure S1B), but also in motor/premotor cortices, in the anterior intraparietal sulcus, and in opercular and insular regions. Subsequent analyses focused exclusively on responsive leads.

Invariant vs. report-related activity

Studies that controlled for the influence of report on sensory awareness emphasized the segregation of sensory- from report-related neural responses, both anatomically^{38,39,51} and temporally.^{9,29} Here, we hypothesized that somatosensory processing of the stimulus, independent of reporting, would be reflected in overlapping activity profiles between NC and RC. Conversely, activities dependent on reporting are expected to exhibit significant differences across NC and RC. To investigate this, we applied a Gaussian mixture model to group all responsive leads based on the correlation and relative magnitude of their responses across NC and RC (see STAR Methods). We identified a cluster of “invariant leads” (ILs, red dots in Figure 2A), characterized by correlated activity and similar response magnitude across NC and RC; a cluster of “report-related leads” (RRLs, cyan dots in Figure 2A), exhibiting no correlation and dramatically stronger response in RC; and a cluster of “mixed leads” (MLs, gray dots in Figure 2A), showing an intermediate behavior. A cross-correlation analysis further emphasized the preserved temporal dynamics across conditions of ILs (Figure S2A), whereas RRLs lacked any shared temporal structure, primarily due to negligible responses in NC. Accordingly, the ILs were found to be significantly activated in both NC and RC, whereas RRLs were substantially activated only in RC (Figure S2B).

As shown in Figures 2B–2E, ILs and RRLs considerably differed in spatial, temporal, and tuning features. The ILs (red dots in Figure 2B) were primarily located in SI, motor/premotor,

(D) Anatomical distribution of responsive leads (see also Figures S1 and S7). In the left hemisphere, we recorded 204 responsive leads out of 2,240 (9.1%); in the right hemisphere, we recorded 83 responsive leads out of 1,115 (7.4%). Abbreviations are as follows: CS, central sulcus; CiS, cingulate sulcus; IPCS, inferior precentral sulcus; STS, superior temporal sulcus; IPS, intraparietal sulcus; SF, sylvian sulcus.

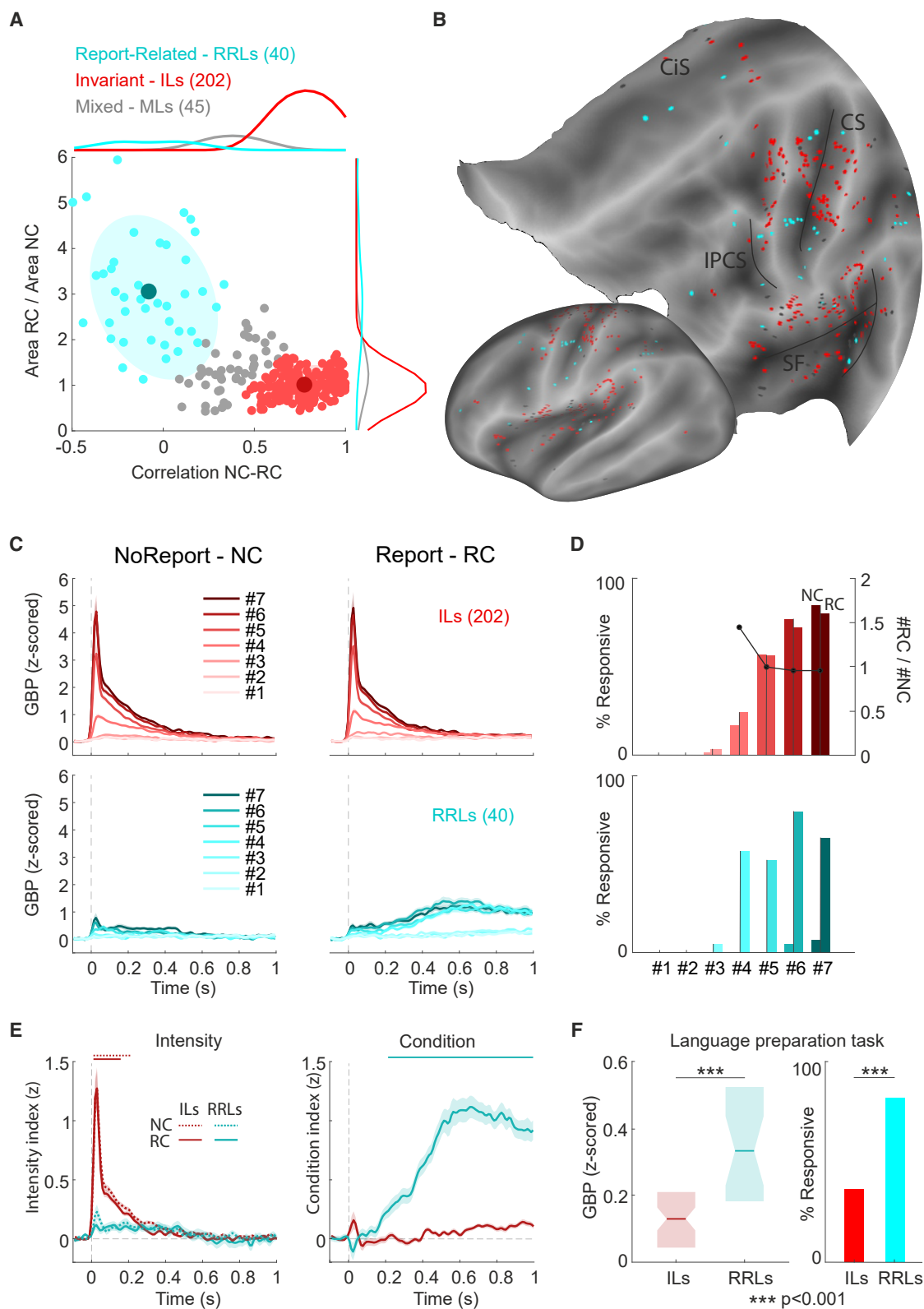


Figure 2. Report-invariant vs. report-related responses

(A) Clustering of responsive leads based on the similarity (correlation on x axis and area-under-the-curve ratio on y axis) of response dynamics across no-report condition and RC. Invariant leads (ILs) exhibit preserved time courses across the two conditions; report-related leads (RRLs) exhibit dramatically different

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parietal cortices, and perisylvian areas of both hemispheres (see Figure S3A), regions known from previous sEEG studies to be activated by contralateral MNS in no-report paradigms.^{5,21} They exhibited, in both NC and RC, an early and high-amplitude activity vanishing around 300–500 ms from the stimulus onset (Figures 2C, top and S3B). Examining the number of active leads across the stimulation intensities, ILs showed a logistic-like trend in both NC and RC (Figure 2D, top), and this aspect further reflects in their exclusive capacity to encode for the stimulation intensity (Figure 2E, left). Conversely, RRLs (cyan dots in Figure 2B) were predominantly located in the left hemisphere (32 left vs. 8 right, $\chi^2 = 3.13$, $p = 0.077$, see Figure S3A), with their distribution mostly concentrated in sectors of motor, premotor, and parietal areas located ventrally relative to the hand hotspot characterized by ILs. The RRLs displayed a ramping, low-amplitude activity profile uniquely for above-threshold stimulations in RC (Figure 2C, bottom), resulting in a responsiveness trend that resembles a step-like function, whereas they remained virtually silent in all NC conditions (Figure 2D, bottom). Consistent with a role in perceptual reporting, RRL responses were similarly present for stimuli delivered to the median nerve ipsilateral to the recording implant, whereas IL responses were largely reduced (Figure S4), in line with previous findings.⁶⁵ Thus, RRLs appeared not to encode the stimulation intensity, although they had a major role in distinguishing RC from NC conditions at stimulations above ST (Figure 2E). Not surprisingly, MLs demonstrated an intermediate behavior (Figure S2).

We finally wondered whether RRLs' activity could be associated with the preparation of the impending verbal report. To test this hypothesis, we retrieved for a subset of our patients the data recorded during the administration of a task requiring the silent reading of short sentences and delayed repetition,^{68,69} examining the responses of RRLs and ILs in the last 500 ms preceding the cue for verbal production. As expected, RRLs were more involved in this phase than ILs in terms of both responsiveness rate and GBP modulation (Figures 2F and S2F).

The stimulation intensity differently modulates phasic and tonic responses

Intracranial gamma responses to above-threshold contralateral MNS in no-RCs are known to cluster into anatomically segregated phasic, prolonged, and tonic patterns.^{5,12} We fitted ILs to these clusters according to the minimum Euclidean distance (see

STAR Methods), revealing an anatomical distribution of the response profiles strongly consistent with previous studies. High-amplitude, short-latency phasic activities are located in SI, primary motor, and parietal regions, prolonged activations are typical of dorsal premotor regions, and low-amplitude, tonic responses mostly characterize perisylvian regions (Figure 3A).

Looking at the response profiles of the three clusters across the stimulation intensities (Figure 3B), one can appreciate that phasic leads increase by ~ 5 times their response amplitude when the intensity rises from ST to MT, whereas prolonged and tonic leads present lower ratios (around 3 and 2, respectively). Phasic leads indeed present the highest tuning for intensity (Figure S5); still, all three clusters confirmed the logistic-like distribution of responsiveness across the seven stimulation intensities, in line with the observation on ILs reported above.

To move beyond the sole responsiveness and investigate whether the different temporal patterns modulate according to the stimulation intensity, we computed the neurometric curve of each individual lead by averaging gamma activity in the 250-ms post-stimulus interval and rescaling values in the range [0–1] across the seven intensities (Figure 3C). This interval was chosen as the minimal time window capturing the dynamical evolution of the three temporal patterns, but overlapping results can be obtained with similar interval choices. Next, we fitted the neurometric curve of each lead with a logistic function to extract the corresponding inflection point values (see STAR Methods). A mixed ANOVA (within factor: condition; between factor: cluster) revealed a significant effect of cluster ($F = 28.3$, $p = 1.6e-11$), with tonic leads exhibiting inflection points at significantly lower intensities relative to both phasic and prolonged leads ($p = 9.6e-10$ and $p = 0.0037$, respectively, Tukey-Kramer correction, Figure 3C). We also found a significant main effect of condition ($F = 8.8$, $p = 0.0034$, Figure 3E), with inflection point values shifted toward lower intensities in RC vs. NC. This effect, quantifiable in terms of subjective sensory and MTs, corresponds to a median increase in the stimulation intensity of 0.29 mA ($Q1 = 0.14$ mA; $Q3 = 0.84$ mA). No cluster $\times$ condition interaction was found ($F = 2.3$, $p = 0.10$).

For a comprehensive view, a continuous map was generated according to the inflection point values of individual leads. The parietal operculum, and especially its posterior part (OP1, OP2, and PFcm^{71–73}), exhibited the inflection point values closest to the ST, a behavior quite expected from NCCs (Figure 3D). On

dynamics and a clear-cut stronger activity in RC; and mixed leads (MLs) exhibit an intermediate behavior (see also Figure S2). Cyan and red ellipses indicate regions containing the 68.8% of data, i.e., the bivariate analog of mean $\pm$ SD intervals. Average silhouette scores^{65–67} are high for ILs (0.49) and lower for RRLs (0.19) and especially MLs (0.09), in line with the intermediate nature of their response.

(B) Anatomical distribution of ILs, RRLs, and MLs, pooled across hemispheres (see also Figure S3).

(C) Average GBP time course of ILs and RRLs ($\sigma = 10$ ms, Gaussian smoothing). Shaded areas represent the standard error. ILs exhibited a consistent time-locked response common to both conditions, whereas RRLs exhibit a prominent activity arising only in the RC (see also Figures S4 and S9).

(D) Percentage of leads significantly activated at each stimulation intensity and condition; the black line indicates the proportion of active leads between RC and NC at each intensity above the ST. The fraction of active ILs is similar across NC and RC, except for a responsiveness increase in RC at the ST. Given that RRLs are almost exclusive for the RC, the same line is not reported for RRLs.

(E) Tuning for intensity and condition in ILs and RRLs. The left panel (intensity) shows the average GBP increment between the perceived intensities, the right panel (condition) shows the average GBP increment between NC and RC across all the above-ST intensities. Shaded areas represent the standard error. Horizontal lines above denote times when the indexes are significantly different between ILs and RRLs (two-tailed t test, $p < 0.01$, at least three consecutive significant bins).

(F) The activity of ILs and RRLs in a control task requiring sentence reading and repetition,^{68,69} conducted on a subsample of eleven patients; compared to ILs ($n = 58$), RRLs ($n = 27$, i.e., the 67.5% or the whole RRLs sample) were significantly more activated (two-tailed t test, $p = 1.7e-5$) and recruited (Fisher's exact test, $p = 1.4e-4$) during a 500-ms preparation phase preceding the verbal production.

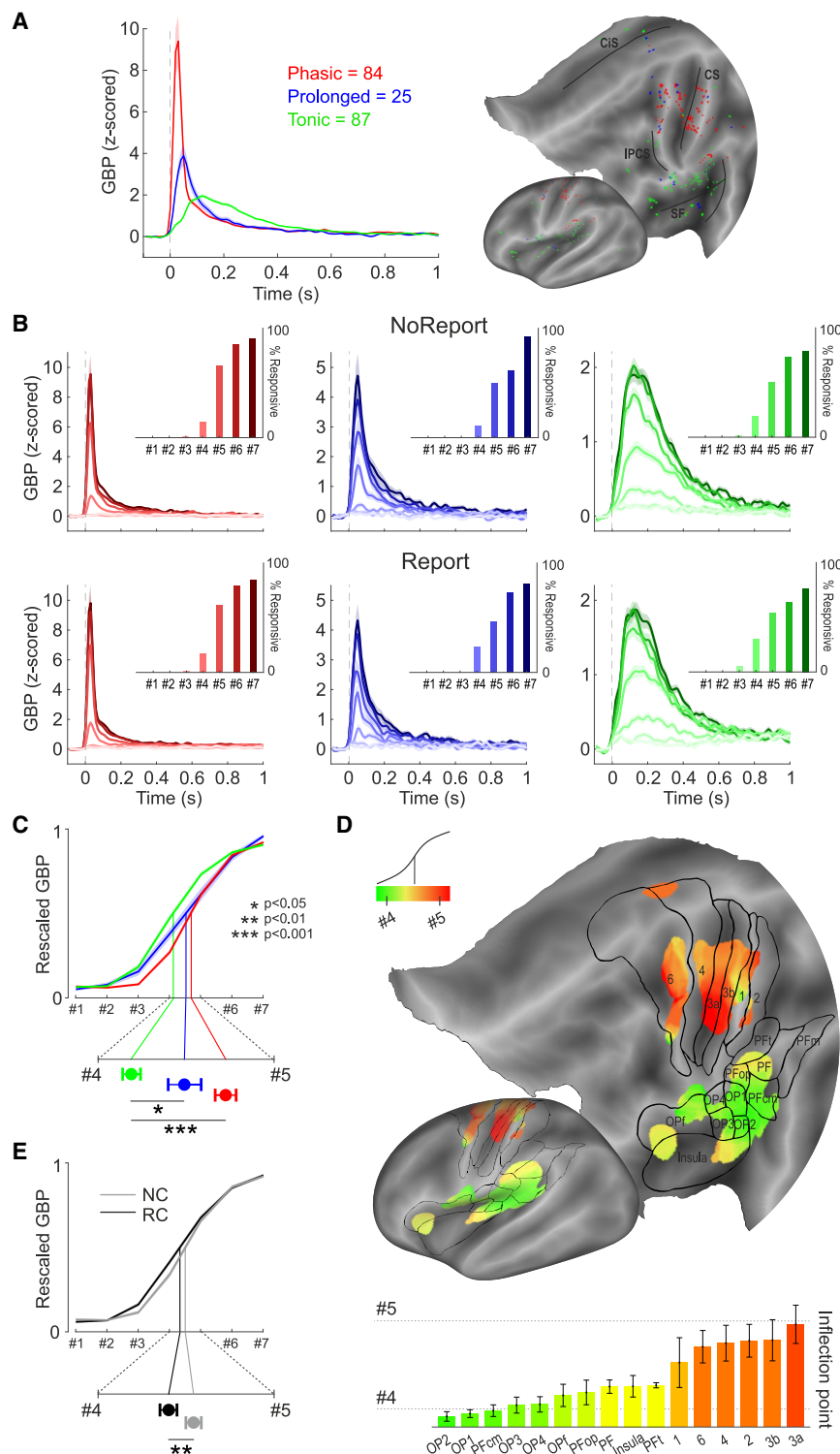


Figure 3. Parietal-opercular neurometric curves inflect at the ST

(A) Left: average profile of phasic, prolonged, and tonic leads (hemispheres pooled, $\sigma = 10$ ms Gaussian smoothing, shaded areas indicate standard error); right: ILs are colored according to the cluster.

(B) Average response for each stimulation intensity in NC (top row) and RC (bottom row). Insets represent the responsiveness of each cluster across the stimulation intensities (see also Figure S5).

(C) Neurometric curves by cluster (average $\pm$ standard error): tonic leads show inflection points at lower intensities relative to phasic and prolonged leads (see also Figure S8).

(D) Top: continuous map of inflection point values, calculated in regions exhibiting responsiveness $>10\%$ (see STAR Methods). Opercular regions—characterized by tonic activity—exhibit inflection points close to the ST, whereas suprasylvian sensorimotor regions—endowed with phasic activity—possess the highest inflection points. Black borders indicate the cytoarchitectonic subdivision—as defined in Amunts et al.⁷⁰—occupied by the inflection point map. Bottom: inflection point values were averaged according to the cytoarchitectonic subdivisions (error bars represent the standard deviation across the nodes of each area). Nomenclature: OP1–4, dorso-caudal, ventro-caudal, ventro-rostral, and dorso-rostral sectors of the parietal operculum, respectively; OPf, frontal operculum; PF, PFm, PFcm, PFT, and PFOp, subdivisions of the inferior parietal cortex.^{71–73} Finally, areas 1, 2, 3a, 3b, 4, and 6 indicate the suprasylvian sensory and motor areas according to the Brodmann parcellation.

(E) Neurometric curves grouped by condition, averaging across clusters (shaded areas represent standard errors): RC presents inflection points at lower intensities than NC (see also Figure S8).

stimulation intensity scale, thus resulting in a lower inflection point. Contrarily, phasic responses continue to increase along with the stimulation intensity, thus making the fitting procedure model the inflection point at a higher intensity.

The encoding properties of tonic vs. phasic responses

Individual tactile perception thresholds change dynamically based on several factors, such as pre-stimulus alpha activity,^{74–76} attention,^{77,78} or the pres-

ence of distractors,^{79,80} so identical peri-threshold stimuli might be consciously perceived in some trials and unperceived in others. Here, stimulation intensities close to the perceptual threshold (75% ST and 110% ST) indeed resulted in variable reports, with the sub-threshold stimulation sometimes being perceived and supra-threshold stimulation occasionally going unnoticed (see light

the contrary, non-perisylvian regions showed higher inflection points. Seemingly controversial, this finding does not indicate a lower sensitivity of suprasylvian regions to somatosensory stimuli (as confirmed by the fact that the earliest responsiveness is found for all clusters at the ST with similar responsiveness rates) but rather that tonic responses saturate earlier along the

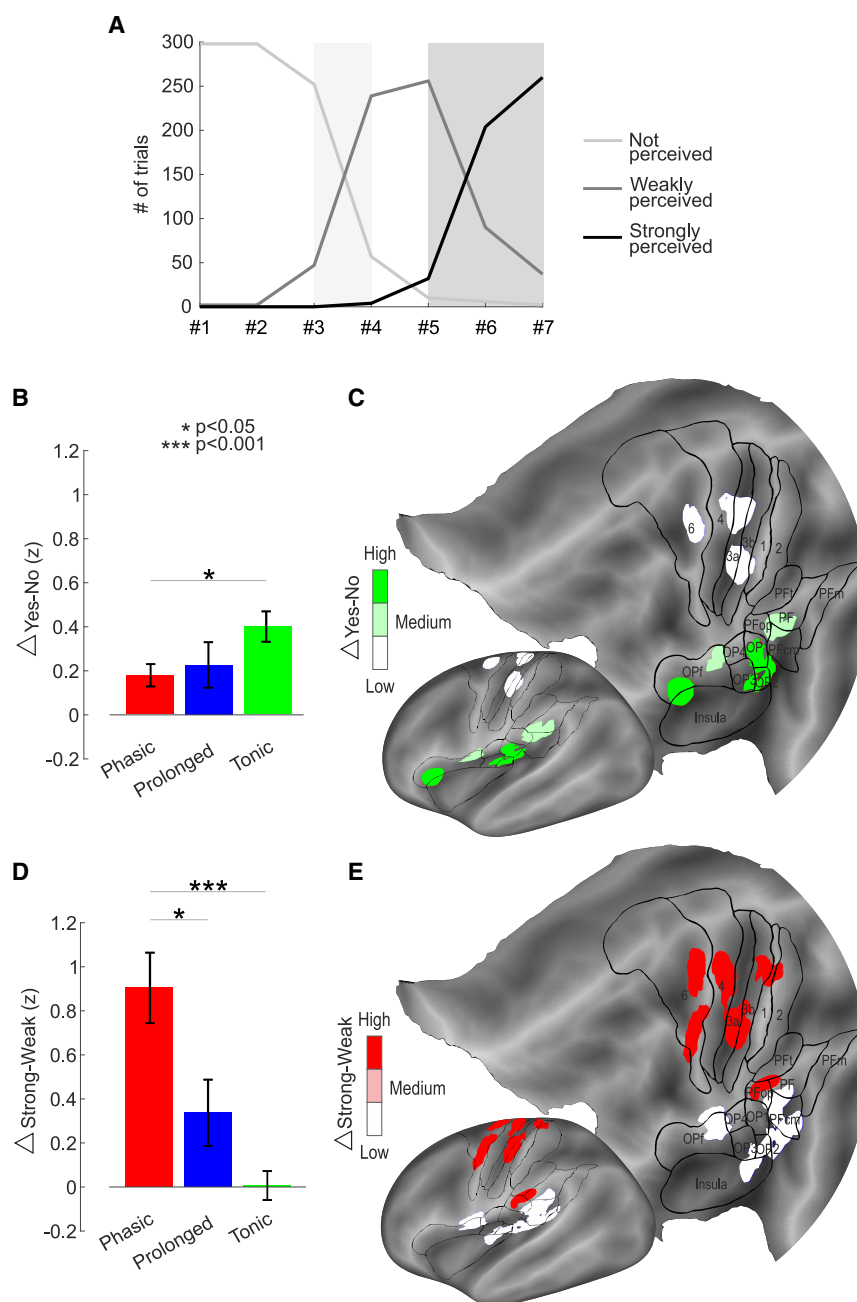


Figure 4. Parieto-occipital activity differentiates between perceived and non-perceived stimuli

(A) Number of trials for each report (unperceived, weakly perceived, and strongly perceived) as a function of the stimulation intensity pooled across all patients. The light gray shaded area denotes peri-ST intensities where the patients' report varies between unperceived and perceived; the dark gray shaded area denotes peri-MT intensities where the patients' report varies between weakly perceived and strongly perceived.

(B) Response modulation between perceived and non-perceived trials across clusters (error bars represent standard errors across leads); tonic activity increases significantly more than phasic activity when stimuli are consciously perceived.

(C) Mapping of modulation between perceived vs. non-perceived trials. Leads were subdivided into low, medium, or high modulation (see STAR Methods), and only regions housing a given class above the 33% (chance level) were displayed. The hubs mostly associated with high modulations were identified in the parietal operculum (OP1, OP2, OP3, and PFcm) and, to a smaller extent, in OPf (see also Figure S6).

(D) Response modulation between strongly and weakly perceived trials across clusters; phasic activity increases significantly more than tonic and prolonged activity when stimuli are perceived as strong vs. weak.

(E) Same as in (C) for the comparison between strongly perceived vs. weakly perceived trials. A reversed pattern emerges here, with SI and motor regions mainly presenting large modulations (see also Figure S6).

gray area in Figure 4A). To evaluate whether variability in the report reflects variability in neural somatosensory responses, we split trials into perceived and non-perceived based on each patient's report and calculated their gamma power difference in the 250 ms following contralateral MNS (see STAR Methods). We found a significantly different response modulation in perceived vs. unperceived trials among the three cortical response clusters (one-way ANOVA, $F = 3.35$, $p = 0.038$), with tonic leads exhibiting a magnification almost double that of other clusters (Figure 4B). Mapping the prevalence of leads with low, medium, or high gamma modulations between perceived and unperceived trials not surprisingly returned a tonic-like topography embedded with high

modulations. However, not all tonic areas had high modulations, which were confined mostly to the parietal operculum (OP1, OP2, OP3, and PFcm) and frontal operculum (OPf) (Figures 4C and S6A; see STAR Methods).

DISCUSSION

The present study, capitalizing on the high spatiotemporal resolution of sEEG recordings, was specifically designed to segregate the cortical neural dynamics underpinning conscious somatosensory experiences from pre- and post-perceptual

modulations. However, not all tonic areas had high modulations, which were confined mostly to the parietal operculum (OP1, OP2, OP3, and PFcm) and frontal operculum (OPf) (Figures 4C and S6A; see STAR Methods).

When applying similar reasoning to high-intensity trials (see dark gray area in Figure 4A) split into those perceived as weak or strong, we found an opposite scenario in which the modulation of phasic leads' activity between strongly and weakly perceived trials significantly

processes. We separated report-related from report-invariant responses, with the latter being confined to the classical somatosensory system.^{1–5,7} Tonic responses in posterior perisylvian regions, particularly the human homolog of the secondary somatosensory area (OP1^{71,81,82}), exhibited the key features of NCC: a report-invariant response to the administered stimuli, an all-or-nothing behavior at the verge of the ST, and the highest magnification between perceived vs. unperceived stimuli. The study provides strong empirical evidence that tactile awareness depends on sustained activations in high-order somatosensory regions, an organizational principle that might extend to other sensory modalities.

Most cortical sites (over 70%) showed similar responses across NC and RC, mainly in peri-central, parietal, opercular, and insular regions. This is consistent with the topographies found in previous no-report studies employing MNS.^{5,12,21,83–85} Regarding timing, report-invariant responses vanished within 300 ms post stimulus, in line with a recent EEG study showing virtually no effect of the report emerging in this temporal range.⁹ The impact of the report was 2-fold. First, it triggered specific responses characterized by late onset (290 ms on average) and mostly localized in the ventral sensorimotor region, typically associated with oro-facial gestures.^{86–90} This activation is consistent with post-perceptual preparation for the upcoming verbal report and matches that observed during verbal preparation in an independent task (see Figure 2F). Second, the report task subtly enhanced report-invariant responsiveness at the verge of the ST, as evidenced by a leftward shift of the neurometric curve between NC and RC. Possibly, this enhancement might reflect top-down attentional modulations driven by saliency or task relevance, factors known to modulate the processing of tactile stimuli.^{91–93}

In contrast with previous studies documenting prefrontal recruitment during report^{29,39,51,94–96} and no-report tasks,^{49,52–54} we found virtually no activations in prefrontal areas, neither report-exclusive nor report-invariant (see also Figure S7). Besides factors related to specific sensory modalities⁹⁷ and the simplicity of the employed stimulation, this discrepancy could have been favored by the sEEG setting, wherein hospitalized patients undergo repeated testing and stimulation protocols. Such habits might reduce the stimulation saliency and associated neural processes^{98–101} relative to typical laboratory contexts. Overall, these results not only de-emphasize the role of prefrontal activations as NCCs in the somatosensory modality but also their involvement in conditions wherein stimuli are consciously perceived while saliency, task,^{40,102} working memory,²⁸ and attention^{93,103} requirements are minimized.

Report-invariant responses were dissected into phasic, prolonged, and tonic dynamics. Phasic activations characterized primary somatomotor areas, prolonged time courses pertained to premotor regions, and tonic responses emerged in posterior insular and opercular regions. These spatiotemporal dynamics have been proposed to fulfill different functional roles within somatosensory processing,^{6,12,21} with early/phasic responses associated with the encoding of the physical stimulus properties and tonic activities advanced as the signature of somatosensory awareness. Phasic activities had the largest sensitivity to the stimulation intensity, with a peak in SI, consistent with previous findings.^{9–11} Importantly, when administered with

identical supra-threshold stimuli, patients reported stronger sensations concomitant with larger phasic components, suggesting that neural processes within SI contribute to the conscious tactile experience. However, SI responses showed limited modulations between noticed and unnoticed perithreshold stimuli, supporting the idea that early processing is not sufficient for conscious detection.^{21,104} Here, it is important to note that the NCCs were originally defined with respect to stimulus detection.²³ However, our experimental design extends beyond the binary distinction between noticed and unnoticed stimuli, considering qualitative aspects of experience, such as variations in perceived intensity. Our findings empirically support the idea that both these aspects are integrated within somatosensory regions, with the neural mechanisms underlying perceptual qualities associated with SI and those related to tactile detection more linked to higher-order somatosensory areas.

The posterior perisylvian region displayed a response profile fulfilling the three criteria of tactile NCCs. First, it significantly responded to tactile stimuli, manifesting the most all-or-nothing neurometric pattern across the ST, in line with Schroder and colleagues.⁸ Second, the caudal parietal operculum exhibited the most pronounced increase in noticed vs. unnoticed perithreshold stimuli, paralleling prior evidence of an elective activity in the secondary somatosensory area following detected vs. missed stimuli.^{7,8,16,95} Third, tactile responses in the caudal parietal operculum remained largely invariant to the report. Together, these points strongly indicate that the NCCs for tactile detection must be found in tonic responses from the parietal operculum. The fact that the criteria for tactile NCCs were matched by neural dynamics with fine-grained temporal and spatial features underscores the importance of sEEG recordings in this research domain.

In agreement with previous reports,^{5,12,21} tonic responses are recorded from most of the perisylvian cortex, a vast territory with functions likely extending beyond tactile stimulus perception. In fact, the awareness-related hotspot in the posterior operculum encompasses five different areas (OP1, OP2, OP3, OP4, and PFcm) with diverse cytoarchitectonic, functional, and connective properties (see Eickhoff et al.¹⁰⁵). Integrating the present findings with previous evidence might help in framing the awareness-related tonic responses in the perisylvian region in a larger context. On the one hand, tactile awareness-related activity in this area may integrate with polymodal inputs, visual inputs impacting tonic activity in OP3,²¹ motor projections reaching OP4,⁸² and limbic processes possibly altering insular tonic activities.¹⁰⁶ On the other hand, the cortical extension of tonic activities is consistent with its potential impact on different brain centers relevant to somatosensation-integrating processes, like haptic exploration,^{107,108} perceptual decision-making,¹⁰⁹ proprioception,¹¹⁰ and perception of social/emotional stimuli.^{111–114} Thus, the perisylvian cortex might represent a platform where different streams (e.g., sensory and motor) affect conscious perception, which, in turn, interacts with related post-perceptual processes.

This study has some limitations. First, we cannot fully delineate the neural mechanisms underlying the subjective experience of a tactile stimulus. Although tactile experience, including perceived intensity, requires activity in SI, stimulus awareness requires

higher-order somatosensory regions. How are these processes coordinated to generate a complete somatosensory experience? Although evidence suggests that tactile awareness emerges through feedback activity from higher order somatosensory areas to SI,^{17,115–117} their dynamic interplay generating the conscious somatosensory experience remains largely unknown.

Second, it is important to note that responses potentially linked to tactile awareness were also observed outside the parietal operculum, such as in the OPf, which showed activity largely report-invariant and greater for hits than misses. Although OPf responses were less tied to ST crossing (Figure 3D), and their awareness-related modulations were lower and more variable than in parietal-opercular areas (Figure S6A), we cannot exclude its involvement in tactile awareness.

Third, because recordings were confined to the cerebral cortex, we cannot rule out the involvement of other brain regions in tactile awareness, such as the thalamus.^{118–120} That said, studies in human and non-human primates have de-emphasized thalamic activity as an NCC, indicating it is not directly linked to tactile detection^{8,16,121,122} but may only appear so when perceptual reports are not properly controlled.⁸

Fourth, anchoring stimulation intensities to both individual sensory and MTs, rather than the sole ST, might appear unnecessary when studying NCCs. Here, we reasoned that the ratio between sensory and MTs varies considerably across individuals, such that using a single anchoring method could have introduced qualitative differences in stimulation across patients. To rule out potential biases from this choice, we tested whether the observed differences across clusters and conditions were affected by the anchoring strategy. The results were consistent across both approaches, but using both thresholds helped in isolating sensory-threshold-related changes from variability in individual MTs (Figure S8). Moreover, when stimulation exceeds the MT, signals reaching the brain reflect a combination of an initial afferent volley directly caused by the stimulation and later refferent signals arising from muscular contraction.^{123,124} Although we found no evidence of refferent signals, even in a region expected to be particularly sensitive to such input (the SI hand hotspot,¹²⁵ Figure S9), the fact that varying intensities elicit both quantitatively and qualitatively different responses is something worth remembering when interpreting our findings.

Despite these limitations, the extensive sampling of the parietal operculum and its consistent display of awareness-related, report-invariant tonic activity support its role as the key region sustaining somatosensory awareness.

In conclusion, our integration of sEEG with a report/no-report paradigm has revealed that somatosensory processing in humans—largely unaffected by the requirement for a report—is composed of a multi-stage organization where tonic responses in the parietal operculum emerge as the key correlates of somatosensory conscious perception. Future single-unit studies will be needed to elucidate which neuronal patterns are responsible, at the micro-scale, for the emergence of the tonic responses observed at the meso-scale. In parallel, given the urge to translate intracranial insights into non-invasive routines, the possibility to simultaneously record sEEG with signals at the macro-scale¹²⁶ could offer

reliable sEEG-informed biomarkers of conscious perception. Lastly, a pivotal question arises regarding whether organizational principles akin to tonic responses extend to other sensory modalities.^{23,127}

RESOURCE AVAILABILITY

Lead contact

Further information and requests for resources should be directed to, and will be fulfilled by, the lead contact, Pietro Avanzini (pietro.avanzini@cnr.it).

Materials availability

The current study has not generated any new material.

Data and code availability

- Data have been deposited and are publicly available as of the date of publication on Open Science Framework: <https://doi.org/10.17605/OSF.IO/7UYC6>. Data will also be made publicly available on EBRAINS: <https://doi.org/10.25493/EF4K-HJH>, as part of a larger dataset. All data are structured in BIDS format (Gorgolewski et al.,¹²⁸ <https://bids.neuroimaging.io/>), following the BIDS proposal for electrophysiological derivatives (BEP021), and are usable with most of the popular sEEG/EEG analysis tools (MNE, EEGLAB, Fieldtrip, etc.). The data are not preprocessed and are only separated into epochs (−0.5 to 1.5 s) relative to MNS. Leads' positions are provided in MNI152 template space (MNI ICBM 2009a Nonlinear Symmetric) following the reconstruction procedure (see below and Avanzini et al.⁵).
- This paper does not report original code.
- Any additional information required to reanalyze the data reported in this paper is available from the lead contact upon request.

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AUTHOR CONTRIBUTIONS

Conceptualization: I.S., M.M., and P.A.; methodology: D.A., M.D.V., I.S., A.P., F.M.Z., E.P.M., and P.A.; investigation: I.S. and F.M.Z.; resources: I.S. and F.M.Z.; data curation: D.A., M.D.V., A.P., and E.P.M.; formal analysis: D.A., M.D.V., and P.A.; writing – original draft: D.A., M.D.V., and P.A.; writing – review and editing: D.A., M.D.V., I.S., A.P., F.T., F.M.Z., S.S., E.P.M., M.M., and P.A.; visualization: D.A.; supervision: I.S., M.M., and P.A.; funding acquisition: M.M. and P.A.

DECLARATION OF INTERESTS

M.M. is co-founder and shareholder of Intrinsic Powers.

STAR★METHODS

Detailed methods are provided in the online version of this paper and include the following:

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SUPPLEMENTAL INFORMATION

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STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Deposited data		
sEEG data	This paper	https://doi.org/10.25493/EF4K-HJH
Experimental models: Organisms/strains		
Human subjects	Human volunteers	N/A
Software and algorithms		
MATLAB 2022a	The Mathworks (https://www.mathworks.com)	SCR_001622
Caret	https://doi.org/10.1016/j.neuroimage.2011.10.077	N/A
Freesurfer	MGH (https://surfer.nmr.mgh.harvard.edu/)	SCR_001847
EEGLAB	https://eeglab.org/	SCR_007292

EXPERIMENTAL MODEL AND STUDY PARTICIPANT DETAILS

Intracranial recording data were collected from 30 patients affected by drug-resistant focal epilepsy (15 male, 15 females, age $m=32$ years, $sd=11$ years) undergoing sEEG implantation of depth electrodes for presurgical evaluations. Seventeen patients had implants on the left hemisphere, 7 on the right, and 6 had bilateral implantation. The NoReport and Report blocks were administered in strict sequence, to exclude possible activity linked to an implicit judgement during the NoReport block.¹²⁹ For the same reason, although patients implanted bilaterally received median nerve stimulation on both hands in consecutive NoReport-Report sessions, only recordings relative to the first session were included in the analysis (left hemisphere, 4 patients; right hemisphere, 2 patients). Before each trial, the experimenter had to manually set the stimulation intensity, an operation that led to variability in the time interval between consecutive stimulations. This ranged typically from 7.0 (average across patients of the 10th percentile) to 11.1 seconds (90th) in NoReport condition, and from 7.3 to 12.4 seconds in Report condition, with an overall minimum of 6.3 seconds across all patients, trials, and conditions.

Implantation sites were selected purely on the basis of seizure semiology, scalp-EEG recordings and neuroimaging examinations (see Table S1). Patients were fully informed regarding the sEEG procedures, and informed consent was obtained. The present study was approved by the Niguarda Ethics Committee (ID 348-24062020). For all patients, neurological examination was unremarkable, and no sensory deficits were presented. Each patient was implanted with a variable number of depth cylindrical electrodes, inserted in the cortical tissue using stereotaxic coordinates. Each cylindrical electrode had a diameter of 0.8 mm and consisted of five to eighteen 2-mm-long contacts (leads), spaced 1.5 mm apart (DIXI Medical, Besancon, France). The procedures for electrode implantation are detailed in Cardinale et al.^{130–132} The sEEG recordings were obtained using a Neurofax EEG-1100 system (Nihon Kohden, Japan) at a sampling rate of 1 kHz. Recordings from all leads were filtered (band-pass: 0.015–500 Hz; notch: 50 Hz). Clinicians selected, for each recorded patient, an intracranial reference based on anatomical and functional criteria, calculated as the average of two adjacent leads exploring white matter. These leads did not show any response to standard clinical stimulations (somatosensory, visual, auditory) and their electrical stimulation did not evoke any sensory and/or motor behavior.

METHOD DETAILS

Stimulations

Patients underwent two blocks (Figure 1B) of peripheral electrical stimulations, delivered by means of a Nihon Kohden - Neuropack M1 MEB-9200 stimulator. In the first block, patients passively received a tactile stimulation of the contralateral median nerve (NoReport Condition, NC). The stimulation was delivered at the wrist contralaterally to the implanted hemisphere using constant-current pulses (0.2-ms duration) at 7 different intensities: 25%, 50%, 75%, and 110% of the sensory threshold (ST); ST+0.6(MT - ST), where MT is the motor threshold (MT); 110% and 130% of the MT (Figure 1A). To determine the MT the stimulation intensity was increased until an observable twitch of the thumb appeared. After determining MT, the stimulation intensity was reduced in steps of 0.1 mA. At each step, the patient was stimulated 10 times. If the patient perceived the stimulation in more than 50% of the trials (i.e., 6 or more), the intensity was further decreased by 0.1 mA. This process continued until the stimulation was perceived in 5 out of 10 trials. The order of stimulation intensity was randomized, and each intensity was delivered 10 times for a total of 70 trials. The second block was

identical to the first, but after each stimulation the patients were asked by the clinician, after a time interval >1 s, to verbally report whether they perceived it and, if so, whether they perceived it as *weak* or *strong* (Report Condition, RC).

Anatomical reconstruction

The individual anatomy of each patient and the coordinates of the leads exploring the grey matter were reconstructed following the procedures detailed in Avanzini et al.,⁵ and individual anatomies were coregistered to a common brain template (FreeSurfer's 164k fsaverage¹³³). All maps were plotted using CARET software¹³⁴ and visualized on either a flat hemisphere surface or an inflated one (see Figure 1).

QUANTIFICATION AND STATISTICAL ANALYSIS

Data processing

First, we selected all the leads located in the grey matter, that were 2550 and 1254 leads in the left and right hemispheres, respectively (see Table S2 for detailed information). Then, to exclude potentially non-physiological responses, we removed all the leads located within or very close to the epileptogenic zone (310 (12.2%) and 139 (11.1%) in the left and right hemispheres, respectively). Data from all the remaining leads (2240 and 1115 leads in the left and right hemisphere, respectively) were decomposed into time-frequency representations using a complex Morlet wavelet decomposition (4 cycles, MATLAB function *timefreq* of EEGLAB suite). Then, gamma-band power (50–150 Hz, GBP) was extracted for all stimulation intensities and conditions in a time window ranging from 100 ms before (baseline period) to 1000 ms after the stimulation and subdivided into 110 non-overlapping 10-ms bins. The resulting single-trial power profiles were z-scored based on baseline values. To reduce the noise introduced by occasional high-amplitude artefacts, we scanned the time bins of all intensities and conditions and blanked the time bins of trials whose power exceeded the across-trials median ± 8 MAD (Median Absolute Deviation).

To assess whether a lead was significantly activated by a given stimulation intensity in a given task, its post-stimulus GBP was compared against baseline using a paired one-tailed t-test ($\alpha=0.05$) corrected by a false discovery rate procedure (Fdr¹³⁵), with the requirement of at least 3 consecutive significant time bins (i.e., 30 ms). All the leads significantly activated by two or more of the five highest intensities (75% and 110% of ST, ST+0.6(MT - ST), 110% and 130% of MT) across conditions (i.e., at least 2 out of 10 intensities $\times$ conditions) were classified as responsive (204/2240=9.1% and 83/1115=7.4% in the left and right hemisphere, respectively). The requirement to be significant to at least two intensities across the two conditions was motivated to reduce the amount of false positive leads. Indeed, some leads were found to be significantly activated by just one of the five strongest intensities across conditions (i.e., 1 out of 10 intensities $\times$ conditions), a significance we considered to be too fragile; to reduce the amount of false positive responses, we tagged those leads as non-responsive (141/2240=6.3% and 70/1115=6.3% in the left and right hemisphere, respectively). A minority of leads significantly activated by the lowest intensities (25% and 50% ST), or whose corresponding average GBP in the 200ms post-stimulation interval exceeded 0.5 z and 50% of the GBP at 130% MT, were tagged as non-responsive, as their response is unlikely to originate from somatosensory processing of the stimulus (28/2240=1.3% and 6/1115=0.5% in the left and right hemisphere, respectively).

Analysis of responsive leads

Invariant leads and Report-Related leads

Responsive leads were clustered based on the similarity of their dynamical response across NC and RC by using two features. For each lead, we trial-averaged the GBP profile in the 1s time interval following median stimulation at 110% MT, separately for NC and RC. We then compared the two profiles by calculating a) their Pearson correlation and b) the ratio between their subtended areas (RC/NC). These two features were calculated for all responsive leads and their distribution was fitted with a Gaussian mixture model (Matlab function: *fitgmdist*) requiring three clusters (see Figure 2A). We chose to fit three clusters instead of two to ensure that leads with intermediate behavior did not contaminate the two primary clusters of interest: Invariant and Report-related leads.

In the cross-correlation analysis (Figure 2B), we took for each lead the post-stimulus 110% MT GBP time-course in NC and compared it with the corresponding GBP in RC at several lags, from -100ms to +100ms. To evaluate the extent to which the clusters defined by the clustering procedure reflected the statistical significance in the two conditions, we assessed the overlap between leads significantly activated by at least one intensity in NC and RC (Figure 2C).

To evaluate the information regarding intensity and condition encoded in ILs and RRLs, we devised two time-resolved indexes for each lead. First, an "Intensity index" was calculated as the mean across the three response differences between the four perceived intensities. Second, a "Condition index" was calculated as the mean across the four response differences between RC and NC across the perceived intensities.

Eleven patients executed a control task where they had to silently read a sentence appearing on a screen after a first cue (Read cue), and they had to repeat it after the appearance of a second cue (Repeat cue), occurring at a fixed interval (1.5 s) after the first (for details see Silva and Citterio⁶⁸ and Abarrategui et al.⁶⁹). For all the leads (N=98) responsive to contralateral MNS, we calculated the GBP in a -0.5 s to +3 s interval relative to the Read cue (the first 0.5 s corresponds to the baseline period). To calculate the GBP, we used the same frequency band used in the somatosensory task (50–150 Hz) and 25 ms time bins. Leads were considered significantly responsive to language preparation if they had a GBP higher than baseline for six consecutive bins (i.e., 150 ms) in the 500 ms time interval preceding verbal production (Fdr-corrected).

Clustering of Invariant Leads

To investigate whether the three main clusters of intracranial response dynamics to median stimulation, i.e., phasic, prolonged, and tonic responses^{5,12} differently encoded the seven intensities, we assigned each IL to the most similar cluster of Avanzini et al.⁵ Specifically, the conditions-averaged GBP time-course of each lead at 110% MT was compared with the centroids obtained in that study at the same intensity, and the cluster having the lowest Euclidean distance was assigned to the lead (leads assigned to the three phasic sub-clusters described in Avanzini et al.⁵ were collapsed into a single phasic cluster).

Fit of neurometric curves

Neurometric curves were calculated for the ILs by taking, for each lead, the trial-averaged GBPs in the 250-ms post-stimulus interval for the seven intensities and by rescaling them between 0 and 1. Each curve was then fitted with a linear and a logistic function, the latter described by the following equation:

$$\sigma(x) = \frac{1}{1 + e^{-(x - x_c)/b}}$$

where x represents the stimulation intensity (ranging from 1 to 7), σ represents the rescaled GBP of one lead, x_c is the intensity value at the inflection point, and b is the slope of the function at x_c . To obtain a single inflection point value for each lead independently from the condition, we pooled trials from the two conditions for each intensity and fitted it with a logistic function as described above.

Correlation with single-trial perceptual report

Trials at stimulation intensities near the sensory threshold (i.e., 75% ST and 110% ST) were split for each patient into perceived/not-perceived trials based on the patient's verbal report. For each lead, we calculated its average GBP across perceived and not-perceived trials at a given intensity (e.g., 75% ST) in the 250 ms following the stimulation and calculated their difference (e.g., $\Delta\text{Yes-No}_{75\%ST}$). The same analysis was applied to the trials at 110% ST to obtain $\Delta\text{Yes-No}_{110\%ST}$, and the two differences were averaged to obtain a unique value describing the response difference between perceived and not-perceived trials for each lead ($\Delta\text{Yes-No}$). The leads of patients who did not perceive all the trials below the sensory threshold (75% ST) and perceived all the trials above the sensory threshold (110% ST) were excluded from this analysis. Finally, the same analysis was applied to high stimulation intensities (i.e., ST+0.6(MT - ST), 110% MT, and 130% MT) to evaluate the response difference between weakly- and strongly-perceived trials.

Functional Maps

The functional maps were computed by assigning to each node of the template cortical mesh a value derived from a weighted mean of all the leads lying within a geodesic distance $d = 10$ mm from the node.⁵ Briefly, the value of each node was derived by a mean of the values of leads located in the surrounding nodes (~ 800), weighted by a logistic function centered on each node (steepness = 2 mm, inflection point = 7.5 mm), such as all leads within 5 mm from a node were maximally weighted, whereas those between 5 and 10 mm were gradually reduced in weight, to limit edge effects. We obtained the following maps:

1. A cortical sampling density map, i.e., the number of explored leads per cm^2 .
2. A responsiveness map, i.e., the number of responsive leads as a percent of the number of explored leads within a distance d from the node.
3. An inflection point map, which represents the average inflection point of fitted neurometric curves across the leads within a disc. Only regions exhibiting a responsiveness higher than 10% were considered.
4. A relative map of the GBP increase in perceived versus non-perceived trials ($\Delta\text{Yes-No}$). This was constructed by classifying leads into *low*, *medium*, or *high* increases based on their value relative to the three tertiles of the total distribution across all leads and then taking the relative percentage of each type of lead within a disc. The resulting relative map was thresholded at 1/3 (which represents the chance level with three classes of equal probability), and clusters of nodes composed of less than 300 nodes were discarded. Only regions exhibiting a responsiveness higher than 10% were considered.
5. A relative map of the GBP increase in strongly-perceived versus weakly-perceived trials ($\Delta\text{Strong-Weak}$), following the same procedure as in 4.

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Supplemental Information

Conscious tactile perception entails distinct neural dynamics within somatosensory areas

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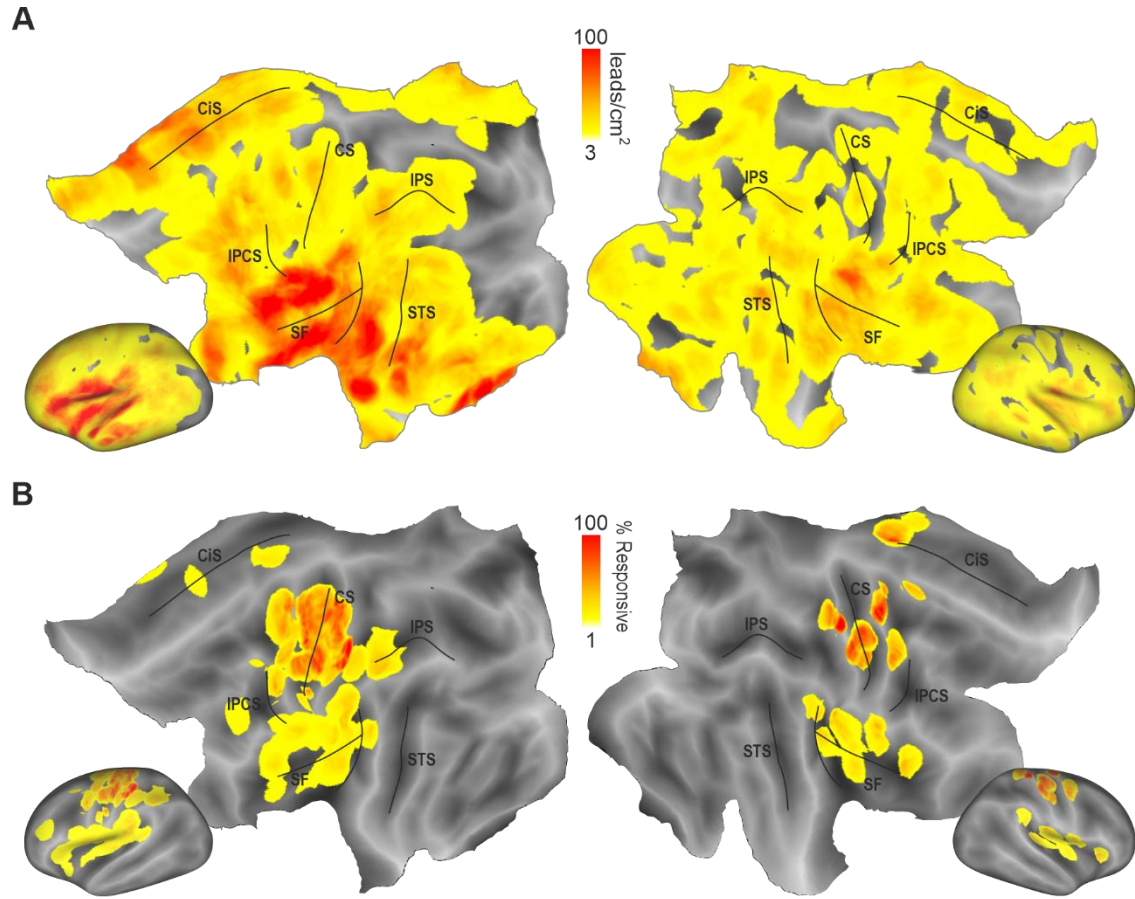


Figure S1. Sampling and responsiveness density maps, Related to Figure 1.

(A) Sampling density map of sEEG recording, i.e., the number of explored leads across all patients per cm² (see Methods). Undersampled regions were limited to frontal and occipital poles and sparse sectors along the cortical crown.

(B) Responsiveness map, i.e., the number of responsive leads within a distance d mm from the node as a percent of the number of explored leads (see Methods).

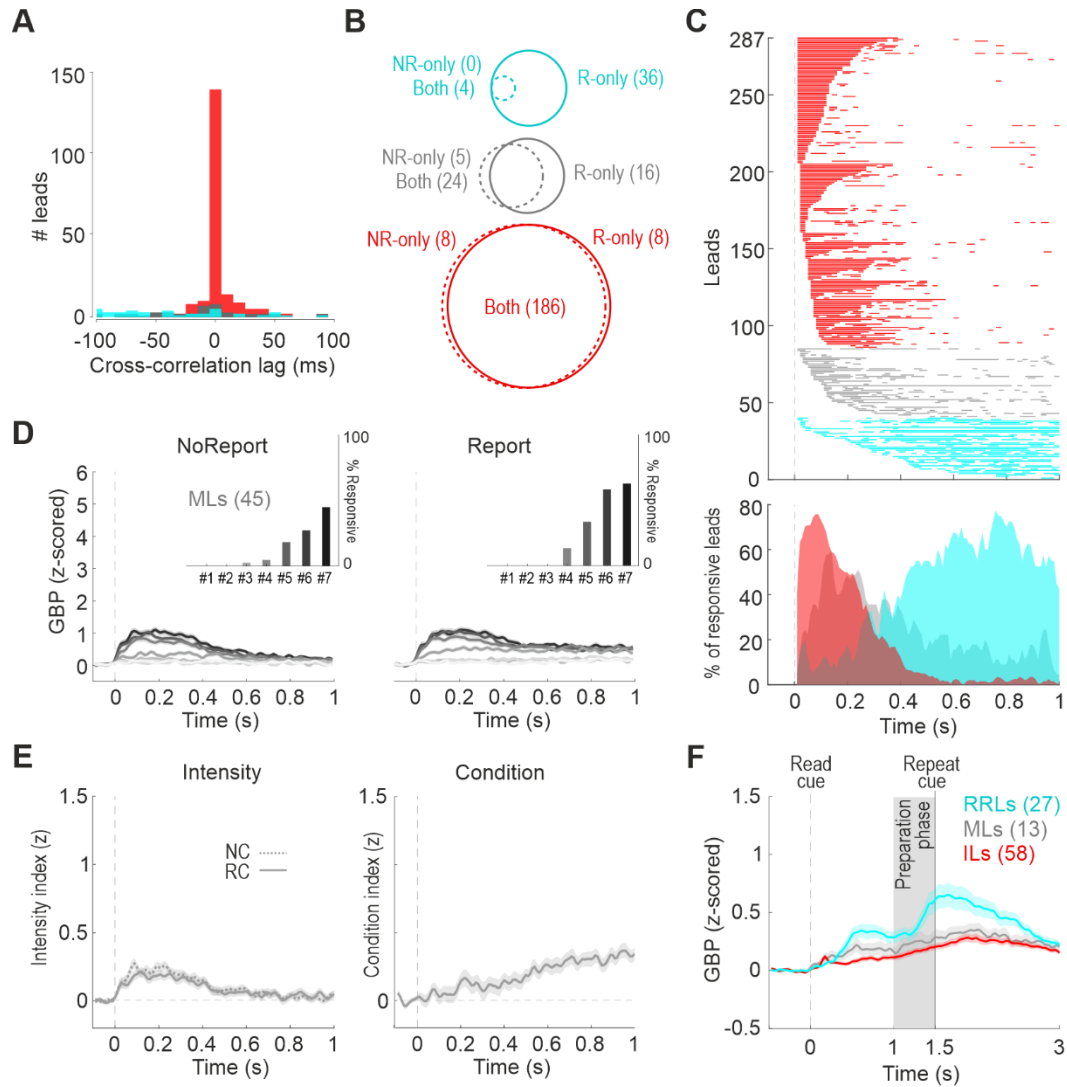


Figure S2. Response features of Invariant, Report-Related and Mixed leads, Related to Figure 2.

(A) Distribution of cross-correlation lags obtained by cross-correlating the GBP time courses of each lead across NC and RC. The majority of ILs exhibit a lag at 0ms, indicating that their responses across NC and RC are temporally overlapping, whereas RRLs show a broad distribution of lags, indicative of the lack of any temporal relationship between the GBP time courses in the two conditions; MLs exhibit an intermediate distribution.

(B) Number of leads significantly activated in NC and RC. ILs are activated in both conditions whereas RRLs are almost exclusively activated in RC; MLs are preferentially activated in RC.

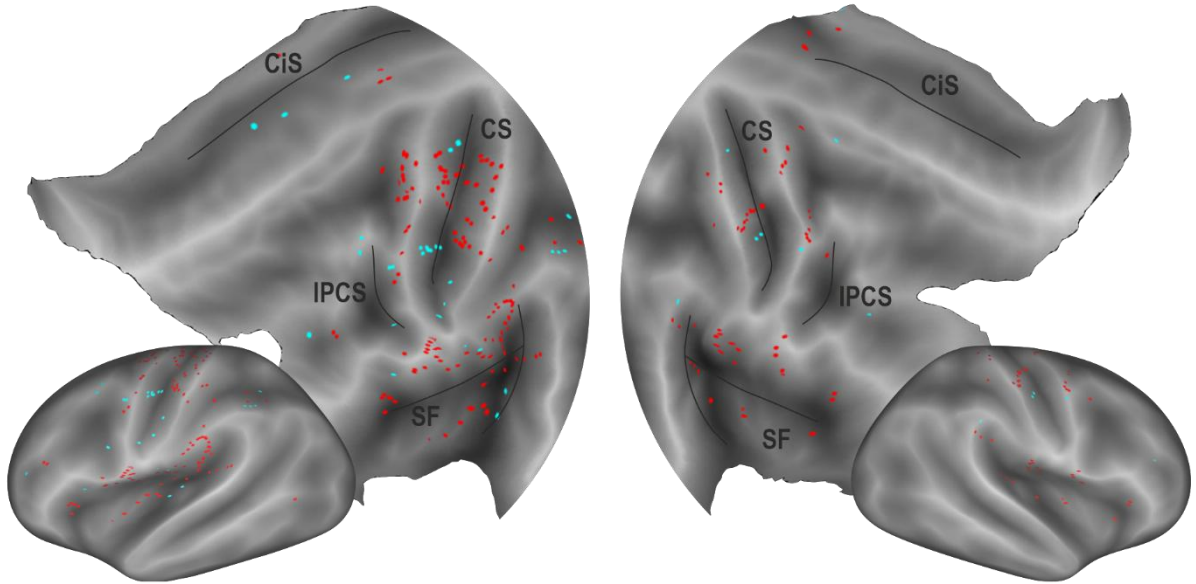
(C) Distribution of significant bins for each responsive lead, pooled across intensities and conditions. ILs' significance is restricted within the first 250-300 ms, whereas RRLs' significance emerges appreciably only after 250-300 ms. Leads are sorted according to their first significant bin within each cluster.

(D) Average GBP time course of MLs ($\sigma=10$ ms Gaussian smoothing). Shaded areas represent the standard error. Insets: percentage of leads significantly activated at each stimulation intensity for the two conditions.

(E) Tuning for intensity and condition in MLs. The left panel (Intensity) shows the average GBP increment between the perceived intensities, the right panel (Condition) shows the average GBP increment between NC and RC across all the above-ST intensities.

(F) Average GBP time course of a subsample of ILs, RRLs, and MLs recorded from eleven patients that executed a language-related task. Briefly, they had to read a sentence appearing on a screen after a first cue (Read cue), and they had to repeat it after the appearance of a second cue (Repeat cue), occurring at a fixed interval (1.5 s) after the first (for task details see^{S1,S2}).

A



B

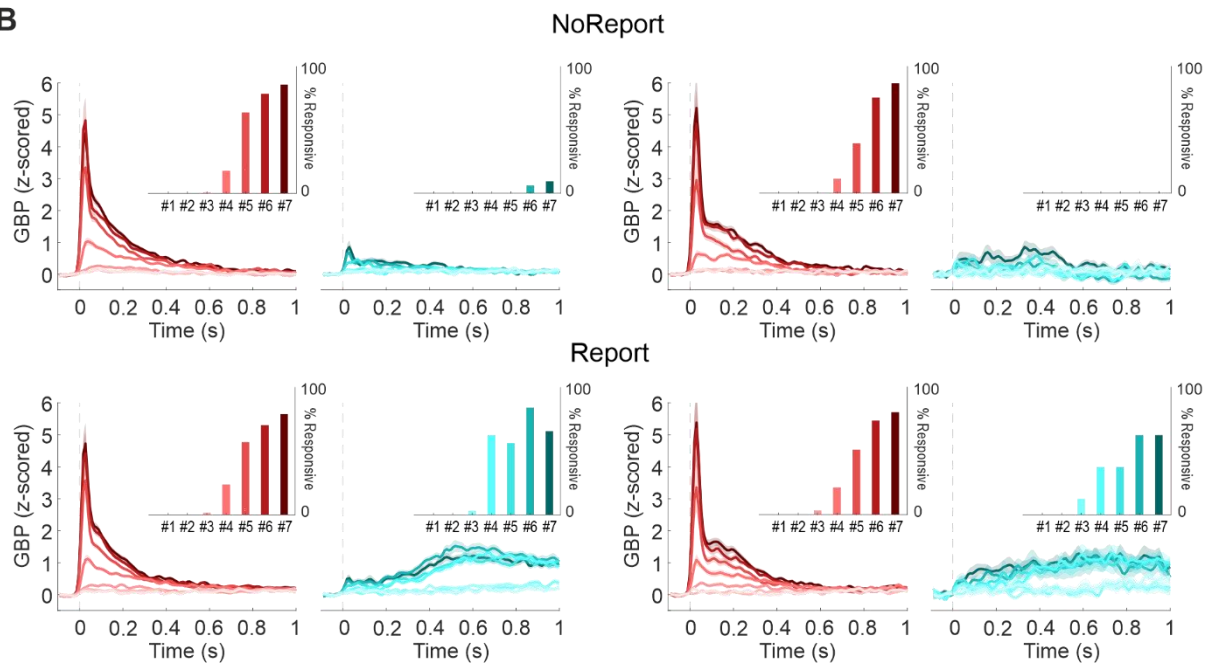


Figure S3. Invariant and Report-related leads across the two hemispheres, Related to Figure 2.

(A) Anatomical distribution of ILs (red dots), RRLs (cyan dots) across the two hemispheres.

(B) Average GBP time course of ILs and RRLs ($\sigma=10\text{ms}$ Gaussian smoothing). Shaded areas represent the standard error. Insets: percentage of leads significantly activated at each stimulation intensity for the two conditions.

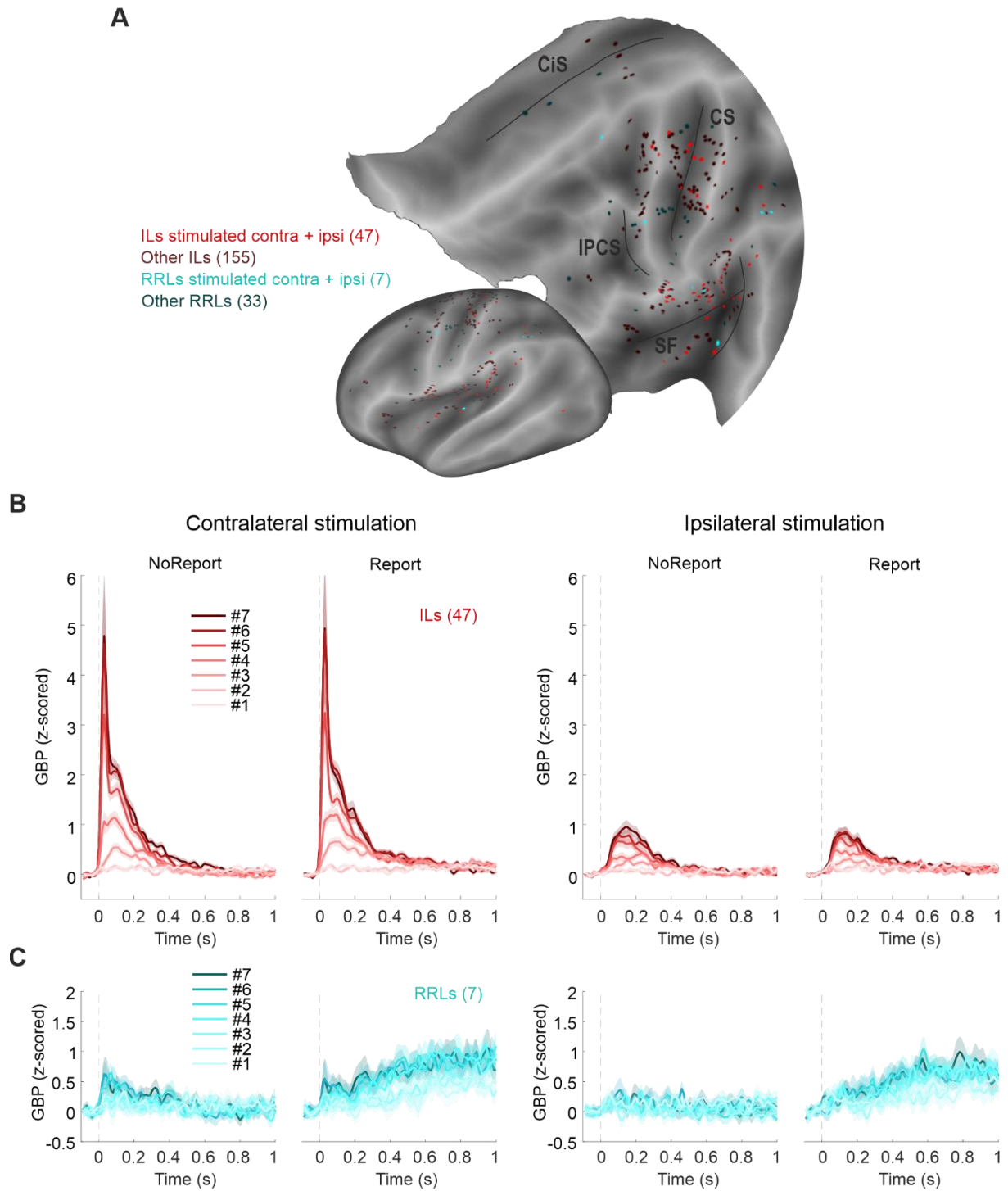


Figure S4. Invariant and Report-related leads during ipsilateral stimulations, Related to Figure 2.

(A) Anatomical distribution of ILs and RRLs recorded during both contralateral and ipsilateral median nerve stimulation relative to the implant (bright red and cyan, respectively), or only during contralateral stimulation (dark red and cyan, respectively). Leads are pooled across hemispheres.

(B) Average GBP time course of ILs following contralateral (left) and ipsilateral (right) median nerve stimulation ($\sigma=10\text{ms}$ Gaussian smoothing).

(C) Average GBP time course of RRLs following contralateral (left) and ipsilateral (right) median nerve stimulation. Conventions as in B.

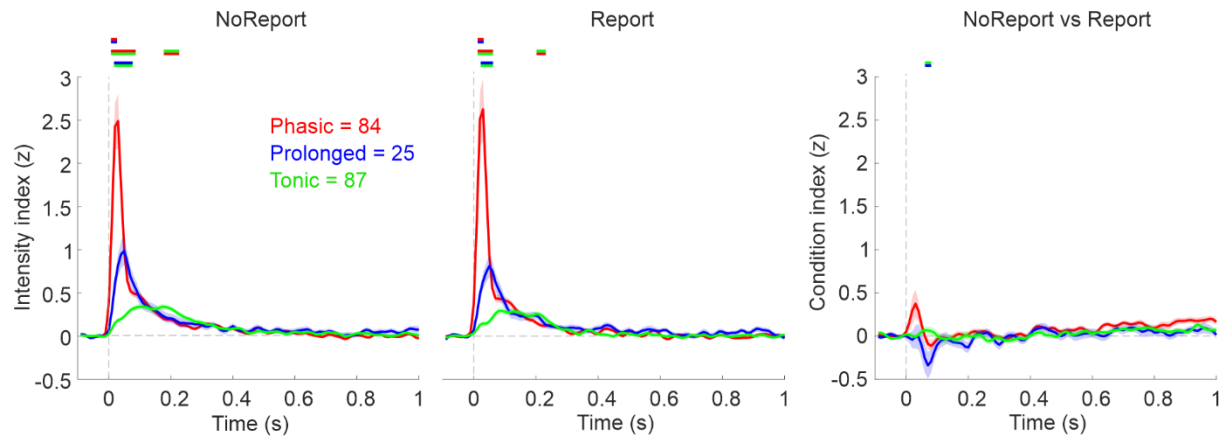


Figure S5. Tuning for intensity and condition across Invariant leads, Related to Figure 3.

Tuning for intensity and condition in phasic, prolonged and tonic leads. The first and second panels from the left show the average GBP increment between the perceived intensities in NC and RC, respectively; the right-most panel shows the average GBP increment between NC and RC across all the above-ST intensities. Double horizontal lines above denote times when the indexes are significantly different across clusters, with the colour of top line indicating the cluster for which the index is higher (two-tailed t-test, $p < 0.01$, at least three consecutive significant bins).

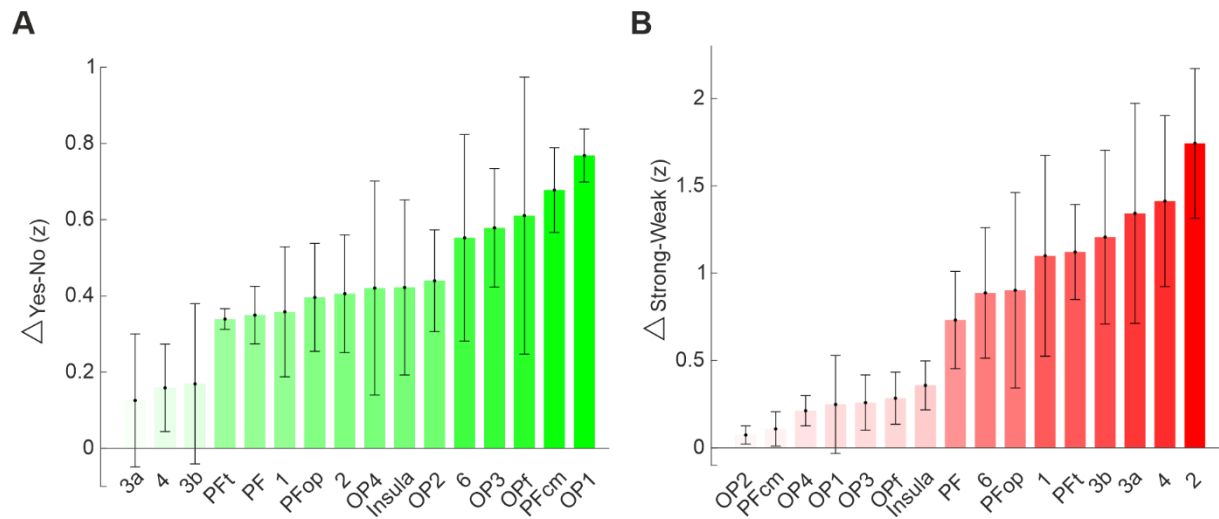


Figure S6. Response modulation based on the patient's report across areas, Related to Figure 4.

(A) Response modulation between perceived and non-perceived trials across the nodes of individual areas. Error bars represent the standard deviation across the nodes of each area. Areal borders are defined according to^{S3}.

(B) Same as in A for the response modulation between strongly-perceived and weakly-perceived trials.

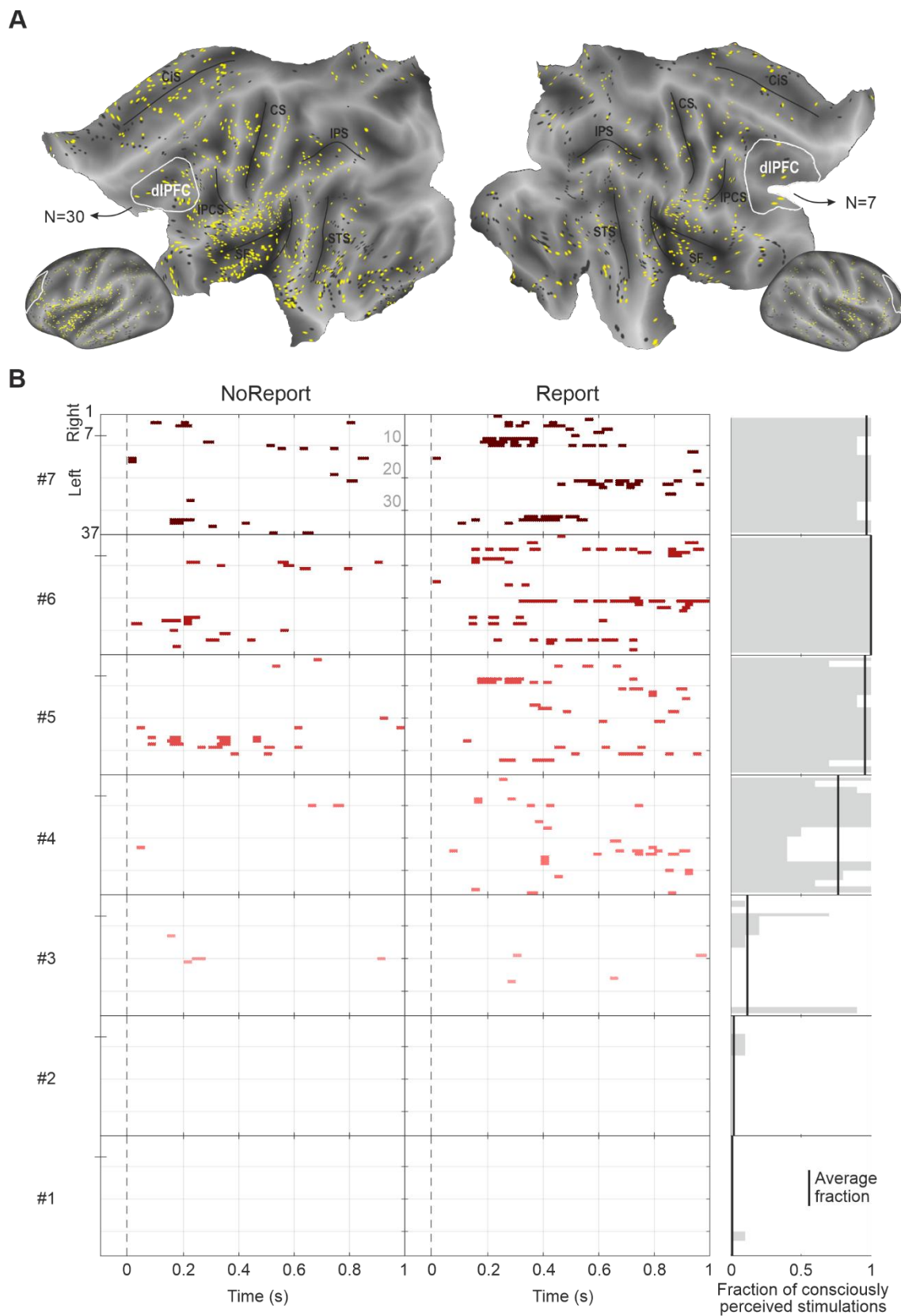


Figure S7. Responses in the dorso-lateral prefrontal cortex are incompatible with a role in sustaining somatosensory awareness, Related to Figure 1.

(A) Anatomical distribution of responsive leads using uncorrected statistical testing, pooling significance across intensities and conditions. White boundaries denote the location of the dorso-lateral prefrontal cortex (dlPFC) according to ^{S4} (areas 46 and 9-46d pooled).

(B) Left: raster of significant bins for dlPFC leads in NC and RC, pooling significance across intensities. Right: fraction of perceived stimuli as a function of the stimulation intensity. Prefrontal leads show weak and inconsistent responses concurrent with evidently perceived stimulations.

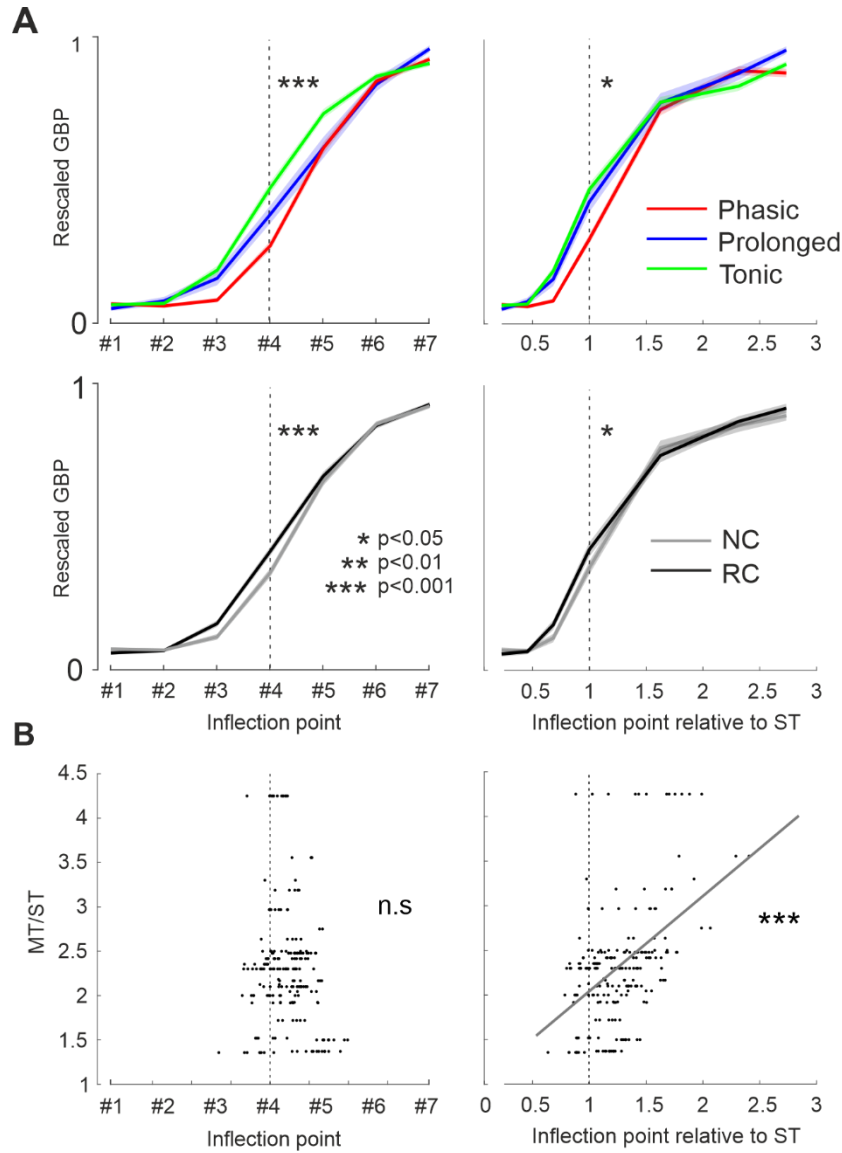


Figure S8. Referencing stimulation intensities only to ST yields results consistent with referencing to both ST and MT, Related to Figure 3.

(A) Neurometric curves of Invariant leads, grouped by cluster and condition. Curves were obtained by referencing stimulation intensities to both ST and MT (left column), or by referencing intensities only to ST (right column). Findings are consistent across both scaling methods; however, referencing to both ST and MT reduces inter-individual variability and emphasize difference within clusters and conditions.

(B) Left: correlation between inflection points (from the ST+MT scale) and patients' MT/ST ratio (Spearman's $Rho = -0.10$, $p = 0.18$). Right: the same analysis for intensities referenced only to ST (Spearman's $Rho = 0.42$, $p = 5.5e-10$). Referencing stimulation intensities to both ST and MT therefore helps isolate sensory threshold-related changes from variability in individual motor thresholds.

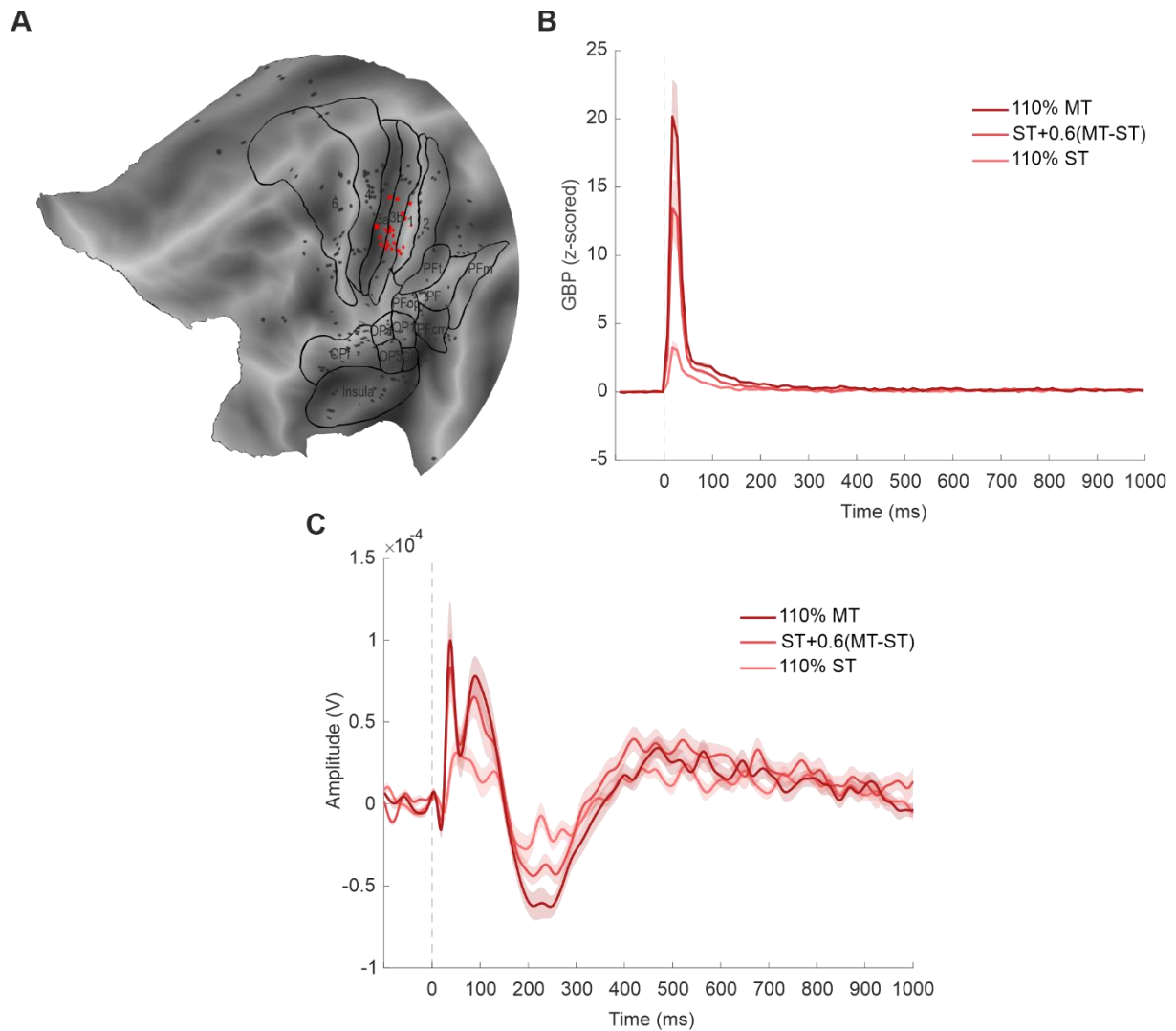


Figure S9. No evident signature of refferent somatosensory signals following stimulation above motor threshold, Related to Figure 2.

(A) Anatomical distribution of the responsive leads located in the hand hotspot of primary somatosensory cortex (SI, red dots, $N = 29$), pooled across hemispheres. All of them were Invariant. For comparison, responsive leads in other regions are shown in grey.

(B) Average gamma band power for SI leads following stimulation intensities below (110% ST, ST+0.6(MT-ST)) and above (110% MT) the motor threshold. NoReport and Report trials were combined. Gamma responses exhibited nearly identical temporal profiles across the three intensities, with differences mainly in response amplitude. Notably, there is no clear evidence of late refferent signals affecting gamma activity above MT.

(C) Same analysis as in B, applied to the event-related potential (ERP). The two highest intensities—one below and one above MT—evoke nearly overlapping ERPs. In contrast, the most pronounced differences occur between the two intensities below MT. Thus, we could not observe in SI ERP data a clear signature of refferent signals following stimulation above MT.

Patient	Gender	Age at implant	Recorded hemisphere	Etiology	MRI	Epileptic zone	Handedness
SUB_01	M	21	Right	Gliososis	Negative	OPf	Right
SUB_02	F	17	Right	/	FTP ischemia	SPL + inferior SMG	Right
SUB_03	F	28	Right	Gliososis	PMG lesion of CFO	AMT + operculo-insular cortex	Right
SUB_04	M	37	Right	FCD Ia of Cingulate cortex	Negative	Anterior SFG	Right
SUB_05	F	31	Right	/	Negative	Multifocal	Right
SUB_06	M	23	Right	/	Negative	AMT + SMG	Left
SUB_07	F	24	Right	Occipital ulegyria	OPT Ulegyria	Temporo-occipital lobes	Right
SUB_08	F	18	Right	/	Negative	Mesio-occipital cortex	Right
SUB_09	M	14	Right	Gliososis	Negative	Frontal lobe	Right
SUB_10	M	38	Left	Gliososis	Negative	OFC + AMT	Right
SUB_11	F	31	Left	/	Negative	AMT	Right
SUB_12	M	44	Left	FCD IIa of parietal operculum	Negative	IPL	Right
SUB_13	M	48	Left	Gliososis	Negative	Temporal lobe	Right
SUB_14	M	16	Left	Gliososis	Negative	Fronto-central lobes	Right
SUB_15	M	22	Left	/	Negative	Parietal lobe	Right
SUB_16	M	48	Left	FCD IIb of Inferior frontal gyrus	Negative	OFC	Right
SUB_17	F	39	Left	/	Negative	AMT	Right
SUB_18	M	47	Left	Gliososis	FI cysts	OFC	Right
SUB_19	M	35	Left	Gliososis	Negative	AMT	Right
SUB_20	F	23	Left	Gliososis	Negative	Bilateral temporal lobes	Right
SUB_21	F	24	Left	/	Negative	Parietal operculum + SMG	Right
SUB_22	M	35	Left	/	Small frontal heterotopic nodule	Bilateral multifocal	Right
SUB_23	M	29	Left	Gliososis	Negative	Central and parietal operculum	Right
SUB_24	F	40	Left	/	Negative	Bilateral temporo-insular cortex	Right
SUB_25	M	37	Left	/	Negative	STG	Right
SUB_26	F	19	Left	Gliososis	Negative	OFC	Right
SUB_27	F	46	Left	/	Negative	AMT + insular cortex	Right
SUB_28	F	41	Left	/	Negative	Temporo-insular cortex	Right
SUB_29	F	24	Left	/	Negative	Fronto pre-central cortex	Right
SUB_30	F	48	Left	/	Negative	AMT	Right

Table S1. Demographic and clinical data of recorded patients. Related to Figure 1.

Abbreviations: FCD, focal cortical dysplasia; OPf, frontal operculum; SMG, supramarginal gyrus; FTP, fronto-temporo-parietal; CFO, centro-frontal-opercular; OPT, opercular parieto-temporal; FI, fronto-insular; SPL, superior parietal lobule; SFG, superior frontal gyrus; AMT, anterior mesial temporal; OFC, orbitofrontal cortex; IPL, inferior parietal lobule; STG, superior temporal gyrus.

Patient	# of recording electrodes	# of recorded leads	# of recorded leads x electrode	# of recorded leads in grey matter	# of recorded leads in grey matter x electrode
SUB_01	17	167	9.8 ± 1.7	121	7.1 ± 2.1
SUB_02	18	215	11.9 ± 2.2	175	9.7 ± 2.1
SUB_03	18	175	9.7 ± 2.3	145	8.1 ± 1.9
SUB_04	15	156	10.4 ± 3.1	104	6.9 ± 1.8
SUB_05	15	153	10.2 ± 2.2	127	8.5 ± 2.3
SUB_06	16	169	10.6 ± 2.1	139	8.7 ± 1.7
SUB_07	18	182	10.1 ± 1.8	142	7.9 ± 1.6
SUB_08	20	187	9.3 ± 1.9	150	7.5 ± 1.8
SUB_09	18	177	9.8 ± 2.3	151	8.4 ± 2.3
SUB_10	18	188	10.4 ± 2.7	131	7.3 ± 1.6
SUB_11	15	148	9.9 ± 2.4	124	8.3 ± 1.9
SUB_12	16	163	10.2 ± 2.1	141	8.8 ± 1.4
SUB_13	17	160	9.4 ± 2.2	141	8.3 ± 2.2
SUB_14	13	145	11.2 ± 1.7	128	9.8 ± 1.8
SUB_15	18	179	9.9 ± 2.0	143	7.9 ± 2.7
SUB_16	16	168	10.5 ± 2.6	128	8.0 ± 2.9
SUB_17	15	178	11.9 ± 2.7	140	9.3 ± 2.2
SUB_18	17	176	10.4 ± 3.1	153	9.0 ± 2.8
SUB_19	14	149	10.6 ± 2.6	115	8.2 ± 1.9
SUB_20	15	141	9.4 ± 2.4	117	7.8 ± 1.5
SUB_21	13	141	10.8 ± 2.3	111	8.5 ± 2.4
SUB_22	12	107	8.9 ± 2.5	89	7.4 ± 1.9
SUB_23	19	192	10.1 ± 2.1	146	7.7 ± 2.4
SUB_24	4	37	9.3 ± 2.6	31	7.8 ± 2.2
SUB_25	16	163	10.2 ± 2.9	97	6.1 ± 1.9
SUB_26	16	173	10.8 ± 2.7	134	8.4 ± 1.7
SUB_27	11	121	11.0 ± 2.8	74	6.7 ± 1.3
SUB_28	16	181	11.3 ± 3.1	152	9.5 ± 3.0
SUB_29	16	184	11.5 ± 2.7	130	8.1 ± 1.9
SUB_30	14	157	11.2 ± 2.4	125	8.9 ± 1.9

Table S2. Recording implants details of individual patients. Related to Figure 1.

Supplementary References

- S1. Silva, G., and Citterio, A. (2017). Hemispheric asymmetries in dorsal language pathway white-matter tracts: A magnetic resonance imaging tractography and functional magnetic resonance imaging study. *Neuroradiol J* 30, 470–476. <https://doi.org/10.1177/1971400917720829>.
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- S3. Amunts, K., Mohlberg, H., Bludau, S., and Zilles, K. (2020). Julich-Brain: A 3D probabilistic atlas of the human brain’s cytoarchitecture. *Science* 369, 988–992. <https://doi.org/10.1126/science.abb4588>.
- S4. Glasser, M.F., Coalson, T.S., Robinson, E.C., Hacker, C.D., Harwell, J., Yacoub, E., Ugurbil, K., Andersson, J., Beckmann, C.F., Jenkinson, M., et al. (2016). A multi-modal parcellation of human cerebral cortex. *Nature* 536, 171–178. <https://doi.org/10.1038/nature18933>.